

MOHAMMED RISHIN POOLAT

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PROFESSIONAL SUMMARY

Software Engineer with a full-stack background spanning user-facing interfaces to backend infrastructure, and a focus on integrating AI and LLMs into production systems. Masters in Advanced Computer Science from the University of Leicester, UK, and a Bachelors in Computer Science, paired with hands-on experience applying machine learning and NLP techniques across relational and NoSQL databases. AWS Certified Cloud Practitioner with a track record of deploying and scaling cloud applications in industry roles.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, HTML/CSS

Frontend & Backend: React.js, Next.js, Node.js, Express.js, ASP.NET Core, FastAPI, TailwindCSS, Socket.io

AI & Machine Learning: LLMs, RAG, Vector Databases, MCP, AI Agents, OpenAI API, Google Gemini API, TensorFlow, DeBERTa, LightGBM, NumPy, Pandas

Databases & Cloud: PostgreSQL, MongoDB, Redis, MySQL, DynamoDB, AWS (EC2, RDS, S3, CloudFront)

DevOps & Tools: Docker, Git, CI/CD, RabbitMQ, BullMQ, Odoo, Puppeteer, Stripe, Linux

EXPERIENCE

Acugence Systems

June 2026 – Present

Software Engineer

Doha, Qatar

- Lead client-facing discovery sessions to gather requirements, running detailed analysis to translate business needs into structured project plans with clearly defined milestones, phases and tasks for delivery.
- Design and build responsive employee-facing portals, handling UI/UX and front-end development alongside the underlying database models, giving client staff self-service access to key workflows and data.
- Develop custom Python modules to extend and customize Odoo ERP across HRMS, time off, payroll and maintenance, tailoring workflows, business logic and reporting to match each client's operations.
- Configured and standardized the team's Claude Code setup, establishing shared conventions and tooling so AI-assisted development could be adopted consistently across engineers.

OffSeason

March 2024 – December 2025

Software Engineer

Leicester, United Kingdom

- Engineered an AI-powered WhatsApp assistant using OpenAI models and the Model Context Protocol (MCP), letting the team query live sales, inventory, and order data across multiple Supabase databases in plain English.
- Built and maintained a full-stack e-commerce platform using Next.js, TypeScript, and Stripe, delivering the full customer journey from browsing and flash sales to secure checkout and order tracking.
- Developed Tracy, an in-house point-of-sale and inventory system serving two retail stores, with barcode checkout, Typesense-powered search, multi-payment processing, and four UK VAT schemes.
- Automated order confirmations, payment-failure alerts, and daily sales-report distribution via WhatsApp and email using Trigger.dev, n8n, and Resend, and built an OpenAI-powered service that converts production errors into detailed GitHub issues.
- Created a suite of web-scraping and data-pipeline services with Puppeteer and Bun to collect product and pricing data from external sources, storing it in the database through scheduled background jobs.

Qburst Technologies

July 2023 – December 2023

Software Engineer

Kerala, India

- Developed a full-stack content management system for internal teams using React, Node.js, and PostgreSQL, providing an intuitive interface and REST APIs to create, organize, and publish content efficiently.

- Built a social media platform enabling users to share images, like, comment, follow, and interact, with relational data in AWS RDS, chat messages in DynamoDB, and media served through S3 and CloudFront.
- Deployed and scaled both applications on AWS using EC2, RDS, and load balancing, with PM2 for process management and monitoring to ensure high availability and reliable performance.

Obraz

Software Engineer

August 2021 – June 2023

Kerala, India

- Built web application features and RESTful APIs using React and Node.js, contributing across the full stack to the company's web platforms, internal tools and client-facing products.
- Trained and fine-tuned machine learning models as part of the company's computer vision and AI projects, working across the model development lifecycle from data to deployment.
- Supported dataset preparation and annotation work to produce clean, well-labeled data for model training, collaborating closely with the team to maintain data quality and consistency across projects.

EDUCATION

Master of Science in Advanced Computer Science

University of Leicester

January 2024 - July 2025

Leicester, United Kingdom

Bachelor of Technology in Computer Science and Engineering

APJ Abdul Kalam Technological University

July 2019 - June 2023

Kerala, India

CERTIFICATIONS

AWS Certified Cloud Practitioner

[Link](#)

Amazon Web Services (AWS)

Issued: September 2024

PROJECTS

AI-Powered RFP Management System | TypeScript, LLM, React, Node.js, PostgreSQL, Docker, RabbitMQ, Redis

[Github](#)

- Built a full-stack procurement platform where an LLM converts requests into structured RFPs, extracts vendor pricing from emails, and scores proposals, with RabbitMQ queuing and Redis caching reads.

Property Transaction Extraction System | Next.js, Express.js, PostgreSQL, Drizzle ORM, Redis, BullMQ

[Github](#)

- Engineered a document pipeline that parses property transaction PDFs, translates and extracts structured fields through queued background jobs, with Redis caching and a searchable dashboard for filtering results.

Expense Tracker with Smart Receipt OCR | .NET 9, React, TypeScript, OCR, Entity Framework, Tesseract.js

[Github](#)

- Implemented a .NET 9 expense tracker on Clean Architecture with a one-way dependency rule, combining client-side Tesseract.js OCR with server-side OCR.space calls to auto-populate receipts, with fallback when OCR fails.

Automated Essay Scoring System | Python, DeBERTa, LightGBM, FastAPI, Next.js, Gemini API

- Combined a DeBERTa transformer ensemble with LightGBM models to grade essays and generate feedback, serving predictions through a FastAPI backend, a Next.js dashboard, and Google Gemini for detailed written feedback.

Penply — Collaborative Study Platform | JavaScript, React, TailwindCSS, Node.js, Express.js, Prisma, Socket.io, AWS S3

[Live](#)

- Created a study platform where students join rooms to chat in real time, share notes and files, and collaborate live, with secure login via Google OAuth and two-factor authentication.